

Algebra 1
Unit 14
Lesson 3 Homework

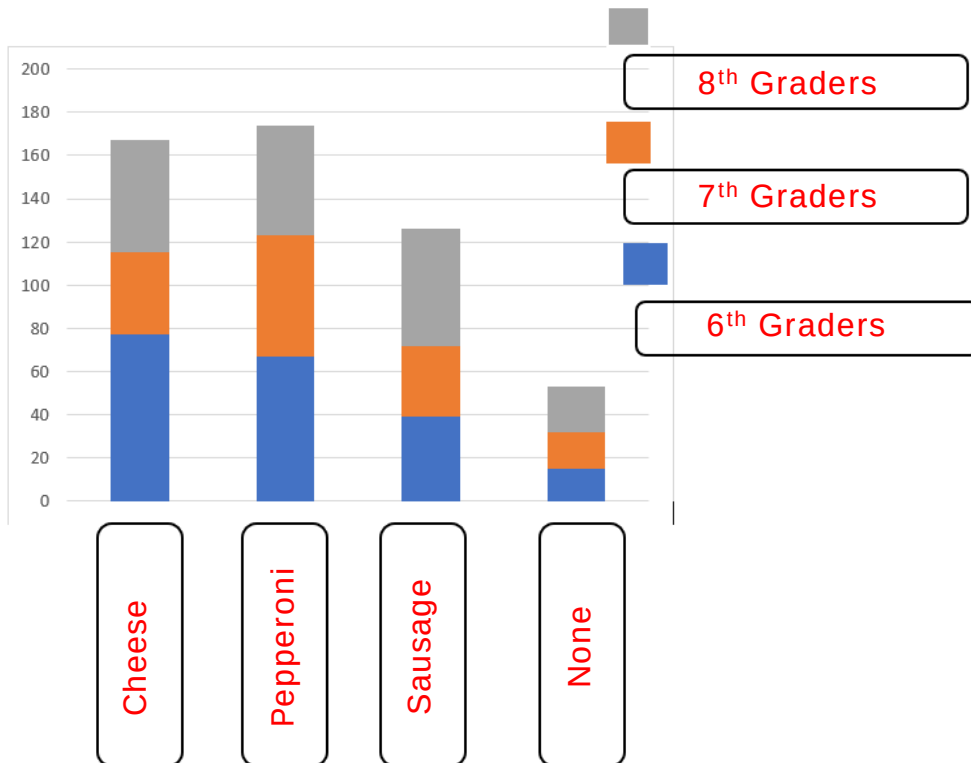
Name Answer Key
Date _____
Period _____

Westside Middle School conducted a school pizza party to celebrate the 100th day of school. Each student was given a choice of cheese, pepperoni, or sausage pizza. Some students did not want pizza. The results are shown in the frequency table below. A segmented bar graph was created with the data from the table. Use the figures to complete questions 1-5.

	6 th	7 th	8 th	Total
Cheese	77	38	52	167
Pepperoni	67	56	51	174
Sausage	39	33	54	126
None	15	17	21	53
Total	198	144	178	520

1. Circle the marginal frequencies in the table.

2. Box the joint frequencies in the table.



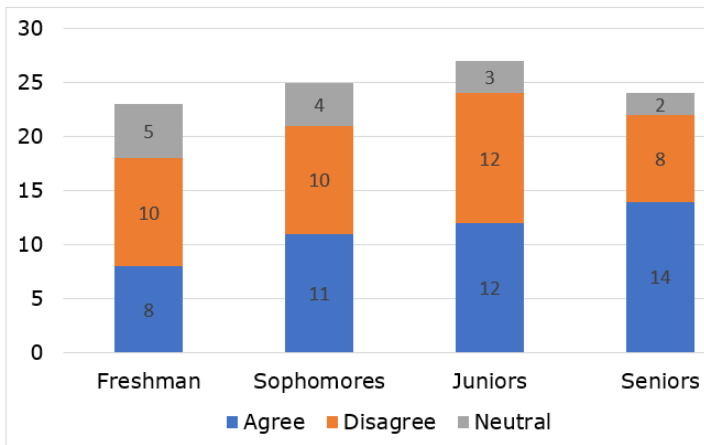
3. Label each marginal frequency in the boxes of the segmented bar graph.

4. Label the value for each joint frequency on the segmented bar graph.

5. Based on the data what type of pizza is the least favorite for each grade level?

6th graders' least favorite is sausage.
7th graders' least favorite is also sausage.
8th graders' least favorite is pepperoni.

The principal at a local high school is considering making uniforms mandatory to improve behavior. The principal polled members from the student council from each grade level to determine if students agree or disagree that uniforms will improve behavior. The results are included on the segmented bar graph below. Use the bar graph to complete questions 6-10.



6. What does the joint frequency value of 14 represent on the bar graph?

The number of seniors that agree that uniforms will improve behavior

7. How many Sophomores disagree that uniforms will improve behavior?

10

8. How many total students polled agree that uniforms will improve behavior?

$8 + 11 + 12 + 14 = 45$ students

9. How many Juniors and Seniors are neutral on the topic of uniforms?

3 Juniors and 2 Seniors

10. If the principal makes her decision based on the data in the segmented bar graph, should she make uniforms mandatory? Explain your reasoning.

Yes, because more students (45) agree than disagree (40).